



SIMATS Online Programming Lab

DS PROGRAMMING



YUGITHA B
192424102RD



QUESTIONS

10 / 10 completed

Linear Search

LINEAR SEARCH

Create a C program to perform Linear Search on an array of 10 integers. The program should allow manual input or random generation of array elements. It must display the first position of the element (1-based indexing), total comparisons made, search status (Found/Not Found), and time taken in milliseconds.

Sample Input:

Enter 1 for manual input, 2 for random: 1

Enter 10 integers:

45 23 67 12 89 34 56 78 90 11

Enter target value: 34

Sample Output:

Target 34 found at position 6

Comparisons made: 6

Status: Found

Time taken: 0.002 ms

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int arr[10], key, i;
5     int pos = -1, comparisons = 0;
6     printf("Enter element to search:");
7     scanf("%d", &key);
8     for(i=0;i<10;i++)
9     {
10         comparisons++;
11         if(arr[i] == key)
12         {
13             pos = i+i;
14             break;
15         }
16     }
17     if(pos != -1)
18         printf("Element found at position %d\n",pos);
19     else
20         printf("Element not found\n");
21     printf("total comparisons made: %d\n". comparisons);
```

OUTPUT

```
Enter element to search:Element not found
total comparisons made: 10
```

INPUT

10 20 30 40 50 60 70 80 90 100
50

© 2026 **SIMATS Online Lab**. All rights reserved.